



UCSC

University of Colombo, Sri Lanka

University of Colombo School of Computing



**DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**

Academic Year 2023— 3rd Year Examination — Semester 6

EN6106 — Emerging Topics in Information Technology

Multiple Choice Question Paper

(1 Hour)

Important Instructions

- The duration of the paper is **1 Hour**.
- The medium of instructions and questions is English.
- This paper has **25 MCQ questions** on **7 pages**. Answer **all** questions.
- Each question will have **5 (five)** choices with **ONE OR MORE** correct answers.
- There will be a penalty for incorrect responses to discourage guessing.
- The mark given for a question will vary from -1 (All the incorrect choices are marked & no correct choices are marked) to +1 (All the correct choices are marked & no incorrect choices are marked). However, **the minimum mark per question would be zero**.
- Answers should be marked on the **special answer sheet** provided.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigilator immediately.
- Mark the correct choices on the question paper first and then transfer them to the given answer sheet which will be machine marked. **Please completely read and follow the instructions given on the other side of the answer sheet before you shade your correct choices.**
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are **not** allowed.
- *All Rights Reserved*. This question paper can NOT be used without proper permission from the University of Colombo School of Computing.

1) When would it be advantageous to use a *monolithic architecture* over a microservices architecture?

- (a) When building a small, simple application.
- (b) When requiring high fault isolation between different parts of the application.
- (c) When the application is expected to have rapid and frequent changes in specific parts.
- (d) When the development team is small and the project timeline is tight.
- (e) When there is a need for a straightforward, unified development and deployment process.

2) What are the characteristics of a *monolithic architecture*?

- (a) Single codebase
- (b) Easy to scale horizontally
- (c) Tight coupling between components
- (d) Centralized logging and monitoring
- (e) Independent deployment of individual components

3) What are the typical challenges of implementing a *microservice architecture*?

- (a) It is easier to implement security policies uniformly across the application
- (b) Increased complexity in managing inter-service communication
- (c) Simplified debugging and testing
- (d) Potential for data consistency issues across services
- (e) Need for robust DevOps practices and tools

4) Which of the following statements about *Docker containers* are correct?

- (a) Containers provide lightweight isolation compared to virtual machines.
- (b) Containers can be easily versioned and rolled back.
- (c) Containers share the host system's kernel.
- (d) Each container includes its own operating system kernel.
- (e) Containers are inherently secure and do not require additional security measures.

5) Which of the following are core components of *Kubernetes architecture*?

- (a) Containers
- (b) Pods
- (c) Nodes
- (d) Services
- (e) ConfigMaps

6) Which of the following transformations might be performed during the *ETL process*?

- (a) Data cleansing
- (b) Data aggregation
- (c) Data encryption
- (d) Data normalization
- (e) Data visualization

7) Select the benefits of *data visualization* from the following list of items.

- (a) Easier identification of trends and patterns
- (b) Reduction in data volume
- (c) Improved decision-making
- (d) Easy way to share information for non-technical audiences
- (e) Enhanced data accuracy

8) What are the benefits of implementing a robust *data engineering pipeline*?

- (a) Real-time data processing and analytics
- (b) Enhanced machine learning model training
- (c) Increased manual data processing
- (d) Efficient handling of large data volumes
- (e) Improved data quality and consistency

9) Which of the following features are commonly associated with *batch processing tools*?

- (a) Real-time data processing
- (b) Scheduled execution of jobs
- (c) High throughput
- (d) Low latency
- (e) Processing large volumes of data

10) Which of the following are common tasks in *computer vision*?

- (a) Image classification
- (b) Object detection
- (c) Connected Graphs
- (d) Semantic segmentation
- (e) Speech recognition

11) What can be considered as sub-domains of *Natural Language Processing*?

- (a) Machine Translations
- (b) Speech Understanding
- (c) Language Generation
- (d) Emotion Detection
- (e) Face Detection

12) What are some benefits of using *specialized AI chips* over *general-purpose CPUs*?

- (a) Higher computational efficiency for AI tasks
- (b) Lower power consumption for AI workloads
- (c) Greater flexibility in programming and task management
- (d) Faster processing speeds for parallel computations
- (e) Reduced cost of hardware development

13) The *systematic approach* to social network analysis can provide numerous benefits, including:

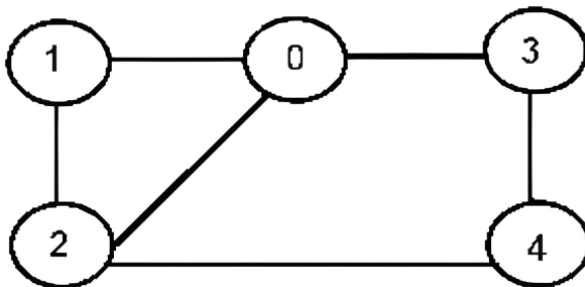
- (a) Objective analysis
- (b) Replicable results.
- (c) help understand local gang issues and relationships
- (d) secure digital forensics
- (e) high data velocity processing

14) Which of the following statements accurately describes *Eigenvector Centrality*?

- I. It measures the influence of a node in a network based on the importance of its neighbors.
- II. It is a measure of the shortest path between two nodes.
- III. It assigns relative scores to all nodes in the network.

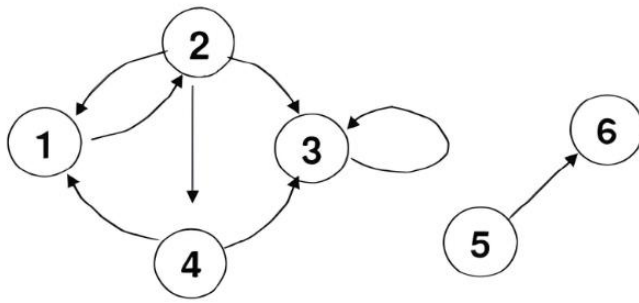
- (a) I only
- (b) I and II only
- (c) I and III only
- (d) I, II, and III are correct
- (e) II Only

15) What kind(s) of *graph data structure* is shown in the following figure?



- (a) Undirected graph
- (b) Biconnected Graph
- (c) Multigraph
- (d) Complete Graph
- (e) Singleton graph

16) Which *adjacency matrix* shows the correct representation of the following graph?



(a)

	1	2	3	4	5	6
1	0	1	0	0	0	0
2	1	0	1	1	0	0
3	0	0	1	0	0	0
4	1	0	0	0	0	0
5	0	0	1	0	0	1
6	0	0	0	0	0	0

(b)

	1	2	3	4	5	6
1	0	1	0	0	0	0
2	1	0	1	1	0	0
3	0	0	1	0	0	0
4	1	0	1	0	0	0
5	0	0	0	0	0	0
6	0	0	0	0	1	0

(c)

	1	2	3	4	5	6
1	0	1	0	0	0	0
2	1	0	1	1	0	0
3	0	0	1	1	0	0
4	1	0	0	0	0	0
5	0	0	0	0	0	1
6	0	0	0	0	0	0

(d)

	1	2	3	4	5	6
1	0	1	0	0	0	0
2	1	0	1	1	0	0
3	0	0	1	0	0	0
4	1	0	1	0	0	0
5	0	0	0	0	0	1
6	0	0	0	0	0	0

(e)

	1	2	3	4	5	6
1	0	1	0	0	0	0
2	1	0	1	1	0	0
3	0	0	1	0	0	0
4	1	0	1	0	0	0
5	0	0	0	0	0	0
6	0	0	0	1	0	0

17) Which of the following techniques are commonly used in *multivariate analysis*?

- (a) [Principal Component Analysis \(PCA\)](#)
- (b) Classification
- (c) [Factor Analysis](#)
- (d) ANOVA (Analysis of Variance)
- (e) [Cluster Analysis](#)

18) Which types of *digital evidence* can digital forensics help to recover and analyze?

- (a) Emails and chat logs
- (b) Financial transaction records
- (c) Physical fingerprints
- (d) Deleted files and data
- (e) Surveillance footage

19) How can digital forensics contribute to solving *cybercrimes*?

- (a) Tracing IP addresses to locate suspects
- (b) Recovering and analyzing digital communication
- (c) Identifying patterns of illegal online activities
- (d) Disguising identities to infiltrate criminal networks
- (e) Providing expert testimony in court

20) What statement(s) is/are truly related to the *Chain of Custody*?

- (a) The process of documenting the handling of evidence from collection to presentation in court
- (b) The physical chain used to secure evidence
- (c) Imprisonment
- (d) The process of encrypting digital evidence
- (e) Ensuring the evidence remains tamper-proof throughout the investigation
